Dawit Bayou Kassie

Senior Software Engineer | Full-Stack, Cloud & DevOps

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PROFESSIONAL SUMMARY

Senior Software Engineer with three + years of experience designing and building production systems, from multi-tenant SaaS platforms to enterprise-grade internal tools and centralized identity infrastructure, currently leading architecture-level design and delivery for enterprise platforms.

Strongest in Node.js, NestJS, Express.js, and TypeScript. Comfortable owning a project end to end - from requirements gathering, BRD/FRS/TDD authorship, and architecture through implementation, deployment, and verification, and equally comfortable working with teams and collaborating.

TECHNICAL SKILLS

- **Backend & APIs**: Node.js, NestJS, Express.js, TypeScript, JavaScript (ES6+); REST & GraphQL API design, WebSockets/Socket.IO;
  Python (Django, FastAPI) and Java (Spring Boot) - working knowledge
- **Architecture & Async Systems**: Microservices, event-driven architecture, Domain-Driven Design, Clean Architecture, multi-tenant SaaS design; Kafka, RabbitMQ, BullMQ, Redis Streams; caching, rate limiting, circuit breakers, horizontal scaling
- **Frontend**: React.js, Next.js, TypeScript, Tailwind CSS, SASS/SCSS; Redux, Zustand, Context API, component-driven and micro-frontend architecture
- **Cloud & DevOps**: AWS (EC2, S3, RDS, Lambda, IAM), Microsoft Azure; Docker, Kubernetes, Terraform, Ansible; GitHub Actions, GitLab CI/CD, Jenkins; Prometheus, Grafana, ELK stack; Linux, Nginx
- **Databases & Search**: PostgreSQL, MySQL, MongoDB, Redis; Sequelize, TypeORM, Prisma; Meilisearch, Elasticsearch, pgvector semantic search
- **Security, Identity & AI**: JWT, OAuth 2.0, SSO, RBAC, Keycloak/Authentik/Zitadel, OWASP practices, AES-256-GCM encryption; LLM integration, RAG, vector search, agentic system design, AI-assisted development with Claude, Codex, and Cursor

PROFESSIONAL EXPERIENCE

Senior Software Engineer --- Kerchanshe Group

2026 - Present Addis Ababa, Ethiopia

- Lead architecture and engineering for enterprise-wide internal platforms supporting operations across one of Ethiopia's largest diversified conglomerates, spanning multiple business units and thousands of users
- Own architecture and delivery across several enterprise projects, including EDAMS, GMS, a retail ERP platform, and centralized SSO infrastructure, coordinating a mix of engineers and AI coding agents
- Architect centralized identity and access management infrastructure, evaluating providers including Keycloak, Authentik, Zitadel, and a custom OIDC build against existing Active Directory infrastructure
- Introduce the verification discipline the team now runs on: every agent prompt requires live HTTP request/response and SQL output as proof before a task is marked done
- Establish CI/CD pipelines, containerized deployment environments, and infrastructure automation practices that reduce manual release overhead and improve deployment reliability
- Own architecture documentation, requirements specs, and executive reporting, BRDs, FRS, TDDs, CEO/CTO-facing presentations, and internal training material

Web Developer --- Edge Communication Technologies P.L.C

Sep 2025 - Oct 2025 Addis Ababa, Ethiopia

- Developed and maintained responsive corporate platforms and internal business applications serving both public-facing and operational needs
- Improved frontend consistency, deployment workflows, and platform performance, reducing page load times and standardizing component architecture
- Collaborated with technical and operational teams to scope requirements and deliver production-ready systems on tight timelines

Full Stack, Cloud & DevOps Engineer --- Freelance

Jan 2023 - Aug 2025 Remote

- Designed, built, and deployed full-stack applications for clients across multiple industries using MERN and PERN architectures, from initial scoping through production launch
- Built backend APIs, authentication systems, and database schemas tailored to each client's actual read/write patterns rather than a one-size-fits-all template
- Implemented DevOps pipelines and containerized environments, standardizing build, test, and deploy stages with Docker, GitHub Actions, and cloud hosting platforms
- Managed AWS and Azure deployments end to end - networking, storage, compute, and IAM policy - configuring security and performance requirements per project

SELECTED PROJECTS

GMS - Enterprise Multi-Tenant Guest Management SaaS Platform

Technologies: *NestJS, Next.js, TypeScript, PostgreSQL, Redis, BullMQ, Docker, Kubernetes, WebSockets, AWS*

- Architected a multi-tenant SaaS platform from the ground up, modular microservices backend with isolated tenant environments, shared infrastructure, and independent per-service scaling
- Shipped a 7-role guest-management platform integrating UniFi Access door hardware, OTP two-factor auth, QR/PIN credentialing, and ID scanning with watchlist screening
- Built the approval/escalation workflow engine and WebSocket layer handling real-time credential delivery, plus centralized auth and tenant lifecycle management reused across the platform
- Ran the QA retest pass, trained internal IT staff, and carried the project through UAT to go-live on AWS with full CI/CD and Kubernetes-based container orchestration

EDAMS - Enterprise Document & Archive Management System

Technologies: *Node.js, NestJS, PostgreSQL, MinIO, Redis, BullMQ, Meilisearch, pgvector, RAG*

- Designed and developed a document and archive management platform using RAG architecture, enabling natural-language retrieval over large internal repositories in place of keyword-only lookup
- Implemented semantic search with vector embeddings and pgvector, and integrated LLM-powered summarization and Q&A so non-technical staff can query large document sets without manual review
- Designed the RBAC and audit layer - vast permissions across many roles and users, HMAC-chained logs, AES-256-GCM encryption at rest, e-signatures, legal hold, and retention scheduling
- Built automated ingestion pipelines for document classification, metadata extraction, and lifecycle access control across multiple document formats and source systems

IDP - Centralized SSO & User Management Platform

Technologies: *NestJS, Node.js, JWT, OAuth 2.0, RBAC, PostgreSQL*

- Built a centralized identity platform providing SSO across a suite of distributed enterprise applications, eliminating isolated authentication and reducing credential management overhead
- Implemented OAuth 2.0 flows and JWT-based session management with configurable token lifetimes, refresh strategies, and audit logging
- Designed tenant-aware RBAC with fine-grained permission scoping, supporting multi-role users across organizational hierarchies

Retail OS ERP - Enterprise Resource Planning Platform

Technologies: *Next.js, Express, PostgreSQL*

- Engineered a retail ERP platform covering point-of-sale, inventory, and back-office operations for a multi-location retail business
- Built inventory and stock-management modules with real-time stock tracking, reorder thresholds, and synchronization across branches
- Developed bookkeeping and financial modules - ledger entries, expense tracking, and financial ratio reporting - plus executive decision-support views summarizing sales, margin, and inventory turnover trends across branches
- Designed secure REST APIs and role-based access controls separating store-level operations from headquarters-level administration

EDUCATION & CERTIFICATIONS

BSc in Electrical & Computer Engineering: Addis Ababa Institute of Technology - 2022

Full Stack & DevOps Engineer Certification: Addis Ababa University - 2025

English --- Fluent · *Amharic* --- Native